

PROGRAMMING METHODS - LABORATORY 5

Program 01

Write and test a program that reads edge definitions of a directed graph from a text file with any name, creates an incidence matrix on their basis, and then determines all neighbors for every vertex.

To create the incidence matrix, use one of the following: a dynamic two-dimensional array, or a vector (`std::vector`). Write the algorithms in functions.

Input

A text file with any name (standard input) has the following format: the first line contains an integer k specifying the number of edges in the graph. Each of the next k lines contains a pair of numbers defining, in order, the number of the source vertex and the number of the target vertex of the edge.

Output

The incidence matrix and all neighbors for each graph vertex.

Program 02

Write and test a program that reads edge definitions of an undirected graph from a text file with any name, creates incidence lists on their basis, and then determines the degrees of all vertices.

To create a single incidence list, use one of the following: structures (singly linked list), classes (singly linked list), or the STL list container (`std::list`). Write the algorithms in functions.

Input

A text file with any name (standard input) has the following format: the first line contains an integer k specifying the number of edges in the graph. Each of the next k lines contains a pair of numbers, i.e. two vertices connected by an edge. The order of the pairs is arbitrary.

Output

The incidence lists and the degrees of all vertices.